

Capturing the Temporal Dynamics of Awe in VR: Using Retrospective Continuous Self-report Method

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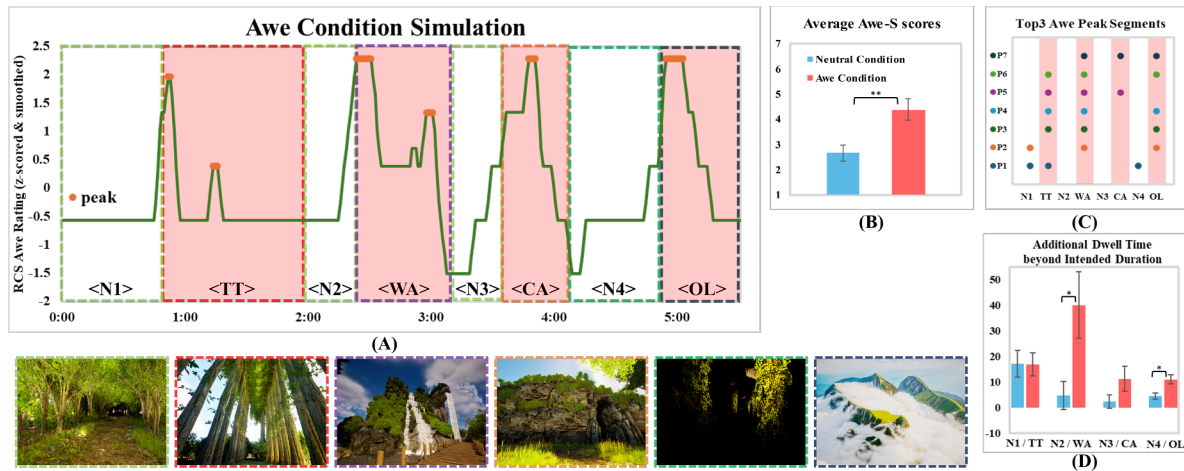


Figure 1: (A) Example of a Retrospective Continuous Self-report (RCS) time series from one participant. Awe ratings (5-point Likert scale; z-scored and smoothed using a moving average with a window size of 3) are shown across the awe condition simulation. Orange dots indicate detected awe peaks, and dashed regions represent simulation segments (N1–N4: neutral; TT: tall trees; WA: waterfall; CA: cave; OL: overlook). (B) Mean post-hoc Awe-S scores comparing neutral and awe conditions. (C) Distribution of the top three awe peaks per participant, illustrating convergence on specific awe cues. (D) Mean additional behavioral dwell time beyond the intended navigation duration for each segment, relative to the preceding neutral segment. Error bars represent the standard error of the mean (* $p < .05$, ** $p < .01$)

ABSTRACT

Research on awe in VR has largely relied on post-hoc self-reports (e.g., Awe-S), limiting insight into its temporal dynamics. This study introduces a Retrospective Continuous Self-report (RCS) method to assess how awe experiences change over time and how people share similar cues that elicit awe. Results showed that awe peaks clustered around specific environmental cues with shared phenomenological features across participants. These moments also involved longer behavioral dwell times, indicating that the RCS effectively captures the temporal unfolding of awe. Overall, the RCS provides a useful complementary tool for studying the dynamic aspects of awe beyond traditional post-hoc surveys.

Index Terms: Virtual Reality, VR, Awe, Emotion Recognition, Temporal Dynamics, Retrospective Continuous Self-report

1 INTRODUCTION

Given the growing interest in awe as a factor in mental well-being, virtual reality (VR) has been increasingly adopted as a reliable tool for inducing awe in controlled laboratory settings [1, 2]. A common feature of awe is that it often intensifies at specific moments during an experience. However, prior studies have primarily relied on post-

hoc self-reports, making it difficult to capture how awe changes over time, which environmental elements contribute most strongly, and how these moments relate to behavior [1, 3, 4]. This limitation presents a challenge, as effectively inducing awe for mental well-being requires a clearer understanding of how and under what conditions it is elicited.

To address this limitation, the present study adopts an exploratory approach by applying a Retrospective Continuous Self-report (RCS) method, in which participants continuously rate their moment-to-moment awe while reviewing a first-person recording of their VR experience [5]. Using this approach, the present study explores the potential of the RCS as a tool to investigate the temporal-dynamics of awe that are difficult to capture with conventional post-hoc measures—specifically, by examining when awe intensifies and how these heightened moments relate to behavioral dwell time.

Building on this approach, the present study addresses three research questions: **RQ1:** Does the awe condition elicit greater awe than the neutral condition in VR simulation? **RQ2:** Do awe-eliciting cues converge across individuals? **RQ3:** Do awe-triggering moments lead to extended behavioral dwell times?

2 METHODOLOGY

2.1 VR Simulation Design

The VR simulations were developed in Unreal Engine 5 (v.5.4.4) and consisted of two conditions: awe and neutral. Both simulations employed a predefined navigation path, within which participants could freely explore. Participants stood and moved using a joystick (maximum speed: 2.1 m/s) with physical body rotation.

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Awe Condition (Forest): The awe condition was designed to elicit awe through perceptual vastness and contrast. The virtual path comprised four neutral segments (N1–N4) interspersed with four awe-inducing segments: Tall Trees (TT), Waterfall (WA), Cave (CA), and Overlook (OL) (see Fig. 1A). The awe segments exhibited pronounced variations in spatial scale (e.g., human-scale vs. massive natural structures) and lighting (e.g., shaded vs. illuminated areas), generating strong perceptual contrasts with the adjacent neutral segments. Immersive environmental sounds (e.g., footsteps, birdsong, waterfalls) were used to enhance presence.

Neutral Condition (Park): The neutral condition served as a control environment, featuring an ordinary park setting composed of sparse distribution of human-scale trees and minimal variation in spatial layout. To maintain a stable baseline experience, environmental changes and visual contrasts were intentionally minimized. Only basic footstep sounds were presented, while other environmental sounds were excluded, in contrast to the awe condition.

2.2 Experimental Setup & Procedure

Participants wore a Varjo XR-4 extended reality headset (3840 × 3744 per eye, 120° FOV) and over-ear headphones to minimize external noise. To validate the awe-inducing simulation, participants experienced both awe and neutral conditions in a within-subjects design, with order counterbalanced. During each VR session, participants' first-person perspectives and in-simulation audio were continuously recorded.

The experiment followed this sequence: simulation, Awe-S questionnaire, RCS task, and semi-structured interview, which was repeated for the second condition. Each simulation lasted ~4–9 minutes, and the full session was completed within one hour.

2.3 Participants

A total of 7 participants with prior VR experience (5 males; mean age = 25.1 ± 1.8 years) were recruited. None of them reported significant cybersickness during the simulation.

2.4 Measures & Analysis

Awe Experience Scale (Awe-S): Awe was assessed using the Awe-S, which measures the overall intensity of awe across six subdimensions on a 7-point Likert scale [4].

RCS: Following each simulation, participants viewed their own first-person recordings accompanied by in-simulation audio and continuously rated their experienced level of awe at temporal resolution of 1 Hz using a 5-point Likert slider [5].

RCS Data Analysis: Raw RCS data were z -score normalized and smoothed with a moving average (window = 3) to reduce noise. Awe peaks were conservatively defined as local maxima with sustained rising and falling slopes lasting ≥ 3 seconds, excluding transient fluctuations (see Fig. 1A). End-of-recording maxima were also treated as peaks when elevated awe persisted for ≥ 3 seconds.

Agreement Index: The agreement index was computed by normalizing inter-participant overlap of the top three segments between theoretical minimum (no overlap) and maximum (complete overlap), yielding a measure of temporal convergence despite individual variability.

Semi-structured Interview: Based on RCS data, semi-structured interviews were conducted after each simulation at awe-peak time points. Participants reflected on perceived triggers, reflected thoughts, and why these moments differed from preceding segments.

3 RESULTS

3.1 RQ1. Verification of Awe Induction

A paired-samples t -test compared Awe-S scores between awe and neutral conditions. Awe was significantly higher in the awe condition ($M = 4.38$, $SD = 1.15$) than in the neutral condition ($M = 2.65$,

$SD = 0.81$), $t(6) = 4.03$, $p < .01$, $d = 1.52$. These results suggest that the developed awe condition effectively elicited stronger awe than the neutral condition (see Fig. 1B).

3.2 RQ2. Convergence of Awe-Eliciting Cues

Based on awe peaks from RCS data, the top three segments per participant were identified by awe intensity and duration. Across participants, six unique segments emerged, yielding an agreement index of 0.83. This value indicates a tendency that VR awe triggers converge on a limited set of environmental cues (see Fig. 1C).

To examine why awe occurred at specific moments, semi-structured interviews were conducted for each peak. Common themes included perceptual vastness (e.g., extreme scale), pronounced environmental contrasts (e.g., narrow-to-open spaces or dark-to-light transitions), and shifts in self-perception (e.g., self-diminishment or release).

Overall, participants showed shared patterns in awe-eliciting cues and phenomenological features, suggesting that VR-induced awe is partly driven by environmental transitions and changes in self-perception.

3.3 RQ3. Behavioral Dwell Time at Awe Cues

Based on RQ2, we examined whether awe-related segments were associated with increased behavioral dwell time. A Wilcoxon signed-rank test compared dwell time at each awe cue segment with the preceding neutral segment. Dwell time was longer for WA ($Z = -2.37$, $p = .018$) and OL ($Z = -2.20$, $p = .028$), whereas the CA segment showed a marginal increase ($Z = -1.86$, $p = .063$) and the TT segment showed no significant difference ($Z = -0.34$, $p = .734$) (see Fig. 1D). These findings suggest that awe-related moments may be associated with extended information-processing demands (need for accommodation) or behavioral inhibition.

4 DISCUSSION

The findings suggest that awe in VR emerges as a structured response shaped by shared environmental triggers, implying that awe-inducing experiences can be intentionally designed. By capturing the temporal dynamics of awe, the RCS method offers preliminary cues about when and how awe tends to intensify, insights that are less accessible through post-hoc questionnaires. This approach shifts the focus from measuring awe to understanding how environmental transitions shape awe responses, offering guidance for designing VR environments that reliably elicit awe and support personalized awe-inducing content. Future work will integrate RCS with physiological signals (e.g., EMG, EEG) to predict awe onset and individual sensitivity to specific cues, aiming to adaptively enhance awe experiences for mental well-being.

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